

# Deployment Strategies

*CSSE6400*

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May 11, 2026

## Branching & Deployment Strategies

- Branching Strategies
  - Deployment Strategies
    - Recreate Deployment
    - Rolling Deployment
    - Blue/Green Deployment
    - Canary Deployment
    - A/B Deployment
    - Shadow Deployment
- Branching strategies help manage development activities.
  - Deployment strategies help manage releasing / deploying systems.

*Definition 0.* Branching

Copying the trunk to allow separate and parallel development.

- Branches deviate from the trunk.
- We will look at a few different branching strategies.

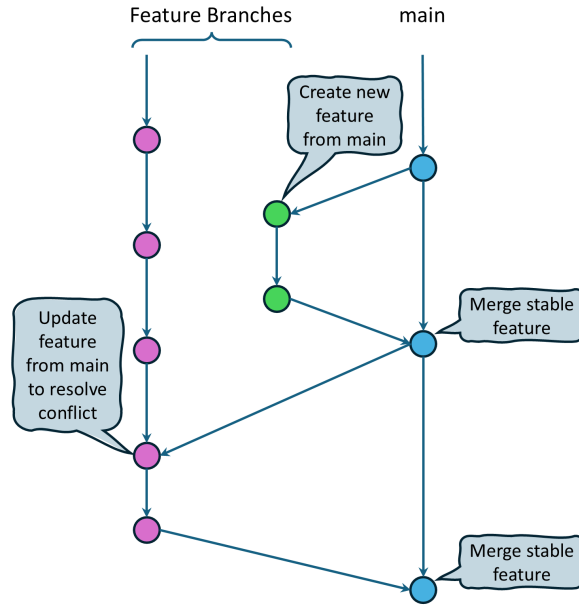
# Branching Strategies

- GitHub Flow
- GitLab Flow
- Release Branches

Branching strategies supporting deployment strategies.

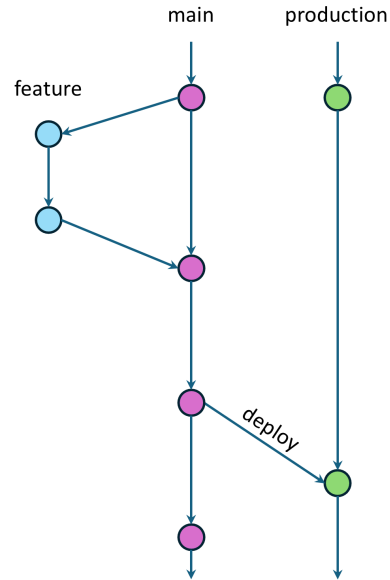
## GitHub Flow <sup>[Haddad, 2022]</sup>

- Main is always *deployable*
- Create branch
- Make changes
- Create pull request
- Resolve issues
- Merge pull request
- Delete branch



- CSSE3200 Feature Branches are a variant of this.
- Supports CI & CD.
- Expects there is a *single* deployable version (e.g. cloud / web systems).

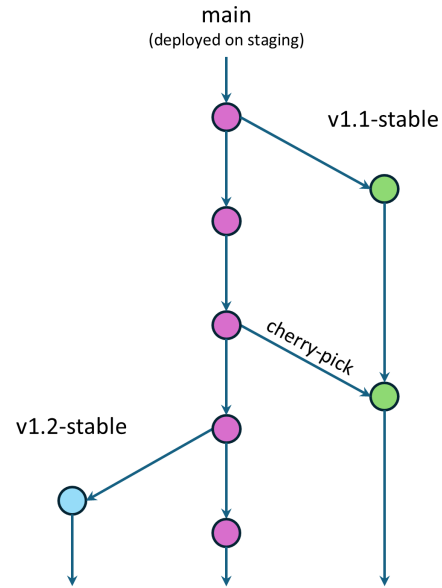
- Supports deployment *windows*
  - Merge to production
  - Deploy when allowed
- Production branch
  - Plus alpha, beta, ...
- Still have
  - Feature branches
  - Pull requests



- Deployment windows examples
  - App store approval
  - Server availability
  - Support availability

## Release Branches [Saavedra, 2023]

- Supports multiple *versions* of system
- Feature development in main
- Released versions are branches
- Bug fixes in main
  - cherry-pick into branches



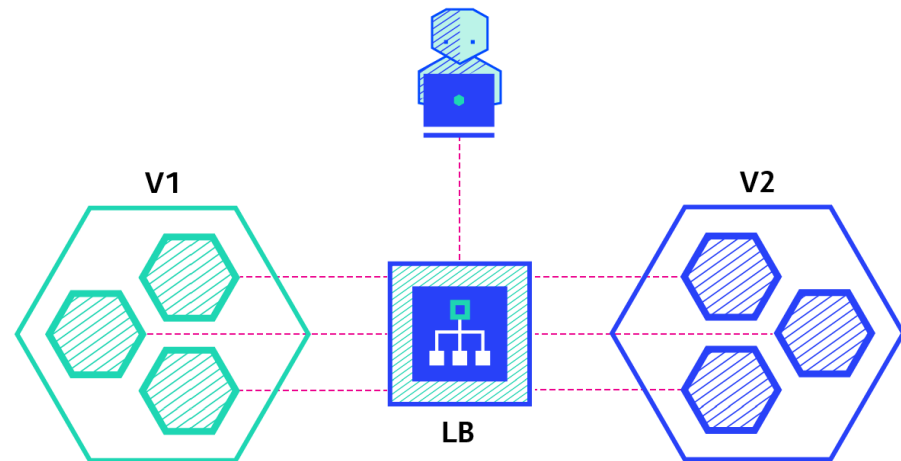
- Cherry-pick: commit is copied from one branch to another, but the branches aren't merged.

*Definition 0.* Deployment Strategy

How a software system is made available to clients.



## Recreate Deployment *[Tremel, 2017]*



- Shutdown version 1.
- Deploy version 2.
- Requires downtime.

## Recreate Deployment

### Pros

- Easy
- Renewed state
  - App reinitialised
  - Persistent storage consistent with system version

### Cons

- Downtime

Renewed state means app is reinitialised and db table structure is consistent with system version.

## Rolling Deployment *[Tremel, 2017]*

- Slowly roll out new version.
- Pool of instances of **V1** behind load balancer.
- Deploy an instance of **V2**.
- Add **V2** instance to pool.
- Remove one **V1** instance from pool.
- Continue until **V2** is fully deployed, replacing **V1**.

## Rolling Deployment

### Pros

- Fairly easy
- Slow release of new version
  - Observe issues
  - Rollback
- Stateful instances can finish gracefully
  - Instance is killed when inactive

### Cons

- Time
- Support multiple APIs
- Support different versions of persistent data structure
- No control over traffic to different versions

## Blue-Green Deployment *[Tremel, 2017]*

- **V2** deployed alongside **V1**, including same number of instances.
- **V2** tested in production environment.
- Load balancer switched to use **V2** instances
- Shutdown **V1** instances.

## Blue-Green Deployment

### Pros

- Instant release of new version
- Fast rollback if necessary
- Only one version ‘live’ at any time
  - No versioning conflicts

### Cons

- Expensive
  - Double the infrastructure
- Stateful instance version switch difficult
  - Can’t kill instance in middle of a transaction

## Canary Deployment *[Tremel, 2017]*

- Gradually shift traffic from **V1** to **V2**.
- Traffic usually split by percent (e.g. 90/10, 80/20, ...).
- Allows a trial deployment to see what happens.

## Canary Deployment

### Pros

- New version released to subset of users
- Can monitor performance and error rates
- Easy and fast rollback

### Cons

- Slow
- Implies poor testing

Canary is commonly used to see if something works or will fail in production.



## A/B Deployment *[Tremel, 2017]*

- Actually it's A/B Testing.
- Both versions are deployed and usage evaluated, usually via analytics.
- Long Term: Deploy version that has best usage result.

## A/B Deployment

### Pros

- Multiple versions run in parallel
- Full control over traffic distribution

### Cons

- Needs intelligent load balancer
- Debugging a version is difficult
  - Need good logs & tools

A/B testing & deployment requires sophisticated infrastructure and analytics to do well.

## Shadow Deployment *[Tremel, 2017]*

- Complex to setup.
- **V2** deployed alongside **V1**.
- All traffic is sent to **V1** & **V2**.
- Tests **V2** ability to handle production load.
- Doesn't impact on production traffic or user experience.
- **V2** rolled out when it demonstrates it is stable.
- Need to manage interactions with external services (e.g. payment gateway).
- When customer checks out their shopping cart, you don't want to send *two* payment requests from **V1** & **V2**.
- Mock external services.
- Persistent data from **V1** (production data) needs to be copied to **V2** when it's deployed as production, with any data migration.

## Shadow Deployment

### Pros

- Performance testing with production traffic
- No impact on users

### Cons

- Expensive
  - Double the infrastructure
- Complex to setup
  - Need mocks for external services

Performance testing may give false confidence.

- It's not user testing.

## Deployment Strategy Options

- Staging or beta testing
  - Recreate or Rolling
- Production (Live)
  - Rolling or Blue/Green
- Uncertain of system stability
  - Canary
- Evaluation
  - A/B or Shadow

There isn't any one perfect deployment strategy.

# Deployment Considerations *[Tremel, 2017]*

| Strategy                                                                                                   | ZERO DOWNTIME | REAL TRAFFIC TESTING | TARGETED USERS | CLOUD COST | ROLLBACK DURATION | NEGATIVE IMPACT ON USER | COMPLEXITY OF SETUP |
|------------------------------------------------------------------------------------------------------------|---------------|----------------------|----------------|------------|-------------------|-------------------------|---------------------|
| <b>RECREATE</b><br>version A is terminated then version B is rolled out                                    | ✗             | ✗                    | ✗              | ■□□        | ■ ■ ■             | ■ ■ ■                   | □ □ □               |
| <b>RAMPED</b><br>version B is slowly rolled out and replacing version A                                    | ✓             | ✗                    | ✗              | ■□□        | ■ ■ □             | ■ □ □                   | ■ □ □               |
| <b>BLUE/GREEN</b><br>version B is released alongside version A, then the traffic is switched to version B  | ✓             | ✗                    | ✗              | ■ ■ ■      | □ □ □             | ■ ■ □                   | ■ ■ □               |
| <b>CANARY</b><br>version B is released to a subset of users, then proceed to a full rollout                | ✓             | ✓                    | ✗              | ■ ■ □      | ■ □ □             | ■ □ □                   | ■ ■ □               |
| <b>A/B TESTING</b><br>version B is released to a subset of users under specific condition                  | ✓             | ✓                    | ✓              | ■ ■ □      | ■ □ □             | ■ □ □                   | ■ ■ ■               |
| <b>SHADOW</b><br>version B receives real world traffic alongside version A and doesn't impact the response | ✓             | ✓                    | ✗              | ■ ■ ■      | □ □ □             | □ □ □                   | ■ ■ ■               |

- Modification of original diagram at <https://thenewstack.io/deployment-strategies/>.
- Cloud Cost for Canary & A/B *increased* to *medium*.
- Rollback Duration for Ramped *decreased* to *medium*.

## References

[Haddad, 2022] Haddad, R. (2022).

What are the best git branching strategies.

[https://faun.dev/c/stories/manuelherrera/  
git-branching-strategies-in-2022/](https://faun.dev/c/stories/manuelherrera/git-branching-strategies-in-2022/).

[Saavedra, 2023] Saavedra, C. (2023).

Combine gitlab flow and gitlab duo for a workflow powerhouse.

<https://about.gitlab.com/blog/2023/07/27/gitlab-flow-duo/>.

[Tremel, 2017] Tremel, E. (2017).

Six strategies for application deployment.

<https://thenewstack.io/deployment-strategies/>.